

Mathematics

Grade 7

Simple Equations

An equation is like a balance.

Whatever you do to one side, you must do to the other.

4.1 What is an Equation?

An equation is a mathematical statement that two expressions are equal.

The equals sign ($=$) shows that the left side equals the right side.

For example, $x + 3 = 7$ is an equation.

4.2 Solving an Equation

To solve an equation, we need to isolate the variable.

For $x + a = b$, subtract a from both sides.

Example: solve $x + 3 = 7$. Subtract 3 from both sides: $x = 4$.

Check: $4 + 3 = 7$. Correct.

4.3 More Equations

Sometimes an equation has a coefficient in front of the variable.

For $2x = 8$, divide both sides by 2 to get $x = 4$.

Practice: solve $3x = 12$.

Chapter 5

Lines and Angles

An angle is formed when two rays meet at a common point.

5.1 Types of Angles

Acute angle: less than 90 degrees.

Right angle: exactly 90 degrees.

Obtuse angle: more than 90 but less than 180 degrees.

5.2 Complementary Angles

Two angles are complementary if their sum is 90 degrees.

Example: 30 and 60 are complementary.